

ANLY 500: Review Material

Week 02: R for Data Analytics

Part 1: Understanding Data Types

```
# Numeric (continuous) variables
age <- 25
height <- 175.5

# Integer variables
year <- 2025L

# Character (string) variables
name <- "John Doe"

# Logical (boolean) variables
is_student <- TRUE

# Factor (categorical) variables
education_level <- factor(c("High School", "Bachelor", "Master", "PhD"))
grade <- factor(c("A", "B", "C", "D", "F"), ordered = TRUE)

# Check variable types
cat("Type of age:", class(age), "\n")

## Type of age: numeric

# Categorical Variables

# 1. Binary variable example: Pass/Fail
pass_fail <- c("Pass", "Fail", "Pass", "Pass", "Fail")
pass_fail_factor <- factor(pass_fail)

cat("Binary variable levels:", levels(pass_fail_factor), "\n")

## Binary variable levels: Fail Pass

# 2. Nominal: Categories without order
departments <- factor(c("HR", "IT", "Finance", "Marketing", "IT", "HR", "Finance"))

cat("Department frequency:\n")

## Department frequency:
table(departments)

## departments
##   Finance      HR      IT Marketing
##         2         2         2         1
```

```

# 3. Ordinal: Categories with logical order
satisfaction <- factor(
  c("Very Unsatisfied", "Unsatisfied", "Neutral", "Satisfied", "Very Satisfied",
    "Satisfied", "Neutral", "Very Satisfied", "Satisfied", "Unsatisfied"),
  levels = c("Very Unsatisfied", "Unsatisfied", "Neutral", "Satisfied", "Very Satisfied"),
  ordered = TRUE
)

cat("Satisfaction levels:\n")

```

```
## Satisfaction levels:
```

```
# Continuous Variables
```

```
# 1. Interval: Equal intervals but no true zero (e.g., temperature in Celsius)
```

```
temperatures <- c(15, 18, 22, 25, 20, 16, 19, 23, 21, 17) # Note: 0°C doesn't mean "no temperature"
```

```
# 2. Ratio: Equal intervals AND true zero (e.g., height, weight, income)
```

```
heights <- c(160, 175, 182, 168, 155, 190, 172, 165, 178, 185)
```

Part 2: Descriptive Analytics

```
# Descriptive analysis summarizes and describes the main features of a dataset using measures of centra
```
